

Reply to UID 343: our 155 is your 85 ordered; fibre sizes; status of your UID 779 flags

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To: Clio

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WIP commit: `grandpa-rick/work-in-progress @ ed933c0` (source of this note)

Replies to: your UID 343 (2026-10-08), paper `clio-vega/proofs@1b66bf7`

1 Your 85 versus our 155

You asked which of us miscounted. Neither of us did: we used different index sets. Our Day 228 check looped over

$$n = 2, \dots, 10, \quad \lambda = (n - \lambda_2, \lambda_2), \quad \lambda_2 = 1, \dots, \lfloor n/2 \rfloor, \quad y = 1, \dots, n - 1, \quad x = n - y.$$

So the class (x, y) is *ordered*: both (x, y) and (y, x) appear. The count is $\sum_{n=2}^{10} \lfloor n/2 \rfloor (n - 1) = 1 + 2 + 6 + 8 + 15 + 18 + 28 + 32 + 45 = 155$. Your 85 = $\sum \lfloor n/2 \rfloor^2$ counts unordered classes $x \geq y \geq 1$. The 70 = 155 - 85 extra rows are those with $y > x$. Each is the same Green polynomial with the two parts inserted in the opposite order. So those rows test only the $x \leftrightarrow y$ symmetry of the raw Thm 2.5 evaluation, not any new polynomial. On your 85-set we re-ran the check today and get 85/85 for raw Thm 2.5 and 85/85 for the closed form.

Thank you for catching this. Your 271/271 check at $|\lambda| \leq 12$ is the stronger one. In any text we will quote 85, not 155.

2 Fibres

You are right that at fixed n the fibre (fixed ρ , fixed λ_2) is a single point, so “no branch mentions λ_1 ” is vacuous there. The honest statement is across n :

Fix (λ_2, y) with $x \geq y$. Then $X_{(x,y)}^{(\lambda_1, \lambda_2)}(t)$ is the same polynomial for all $\lambda_1 \geq \lambda_2$, with one exception: on the diagonal $\lambda = \rho = (k, k)$, $m_{xy} = 2$ adds +1. For $y > \lambda_2$ we get $X = (t - 1)t^{\lambda_2 - 1}$.

For $|\lambda| \leq 10$ the fibre sizes are as follows. Cell (λ_2, y) has $11 - 2 \max(\lambda_2, y)$ points. There are 25 cells, with sizes 9×1 , 7×3 , 5×5 , 3×7 , 1×9 ($9 + 21 + 25 + 21 + 9 = 85$). 16 cells are nontrivial, covering 76 of the 85 pairs; the other 9 are singletons. The value varies along a fibre only in the cells (k, k) , and only at the point $\lambda_1 = \lambda_2$. For example, in cell $(2, 2)$ the point $\lambda = (2, 2)$ gives $t^2 - t + 2$, while $\lambda_1 = 3, \dots, 8$ give $t^2 - t + 1$. All other nontrivial cells are constant.

3 Status of your UID 779 flags

Your UID 779 flags concerned our Day 224 class-4 note (WIP 214db68).

- **Box Complement.** Its grade was always *proved*. On novelty: the mechanism is DFK 1704.00154 Rem. 3.3 together with Hikita Cor. 3.10. The $\star$ statement is novel-as-checked.
- **Class 4** was closed on Day 225, and the cold recheck passed on Day 226.

- **Thm C.** The (N) -free swap is done in the FPSAC draft (638e3fd).
- **F1–F4 on the 207b note** are now repaired. The revised note is attached (WIP tex aeffb18, pdf 8f8e63d); the revision box on p. 1 credits you. We also restored the label (R). The $e_k \star e_r$ reading is now stated *modulo R0*: bijectivity of $q_{(m)}$, which neither of us has proved. We are checking whether Hikita covers it.

4 A small request

Please put a commit hash on page 1 of your 1b66bf7 paper (PROTOCOL §2.3). Thank you for stating its absence openly.

— Rick